

FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Department of Computer Engineering

Academic Year 2025-2026

Rubrics for Lab Experiments

Class: B.E. Computer
Engineering

Subject Name: SMA Lab

Semster: VIII

Subject Code: CSDL8023

Practical No.	2
Title:	Collect Youtube Comments and Perform Sentiment Analysis on it
Date of Performance:	26-11-2025
Roll No.	9913
Student	Mark Lopes

Evaluation

Performance Indicator	Below Average	Average	Good	Excellent	Marks
On Time Submission (2)	Not submitted(0)	Submitted after deadline (1)	Early or on time submission (2)	—	
Test cases and output (4)	Incorrect output (1)	The expected output is verified only a for few test cases (2)	The expected output is Verified for all test cases but is not presentable (3)	Expected output is obtained for all test cases. Presentable and easy to follow (4)	
Coding efficiency (2)	The code is not structured at all (0)	The code is structured but not efficient (1)	The code is Structured and efficient. (2)	-	
Knowledge(2)	Basic concepts not clear (0)	Understood the basic concepts (1)	Could explain the concept with suitable example (1.5)	Could relate the theory with real world application(2)	
Total					

Signature of the Teacher:



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Experiment No: 2

Aim: Data Collection-Select the social media platforms of your choice (Twitter, Facebook, LinkedIn, Youtube, Web blogs etc.), connect to and capture social media data for business (scraping, crawling, parsing).

Objective: To capture social media data and perform sentiment analysis from business perspective.

Lab outcomes:

At the end of this lab session, students will be able to...

1. Acquire hands on skills needed to work with social media data.

Theory:

- Crawling: Involves going through specific websites and related links, more like going through a collection of things and inspecting them. This is the first stage of scraping and Parsing. basically, visiting and going through a site could be termed crawling that is if being done by a bot; web crawler.
- Scraping: It is a form of copying, in which specific data is gathered and copied from the *web*, typically into a specified storage location.
- Parsing: This involves breaking down the above scraped data into smaller bits of it, this is to aid understanding of the scraped data. This stage is employed in various fields of data extraction and mining.

Steps

● **Step1:**

- Get API Key - Go to the Google Cloud console.

(<https://console.cloud.google.com/>) and sign in with your Google Account

- Click the project drop-down menu in the top bar and select or create the project you want to use.
- Click the hamburger menu in the top left and select APIs & Services >Select Youtube Data V3 API> Credentials.
- Click the Create credentials button and select API key.



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- The API key will be displayed in a pop-up window. You can click the RESTRICT KEY button to restrict the API keys usage, such as by IP address or referrer.
- Click the COPY button to copy the API key to your clipboard.
- You can use the API key in your application to access the Google Cloud APIs. Be sure to keep the API key confidential, as it can be used to access your Google Cloud resources.
- Step2
- Search Youtube Video for scraping comments.
- Copy the Youtube Video Id

Student's Task

1. Select youtube video link of your choice and write a code to print comments of that video.
2. Write a code to perform the sentiment analysis of the comments of the selected video using Textblob and Vader.
3. Write your remarks on the results of analysis done in qno 2.

TextBlob and VADER produced slightly different sentiment results. TextBlob is more influenced by longer sentences and often labels many comments as neutral. In contrast, VADER performs better for social media content because it can interpret emojis, slang, capitalization, and intensifiers (such as “VERY good”).

In my analysis, most comments were either positive or neutral, and VADER captured the tone more accurately. For business applications, VADER appears to be the more suitable option since it is more reliable for short, informal YouTube comments.

4. **Article discussion**



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Title: Sentiment analysis for mining texts and social networks data: Methods and tools.

Zucco, Chiara, et al. "Sentiment analysis for mining texts and social networks data: Methods and tools." *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 10.1 (2020): e1333.

Aim: Collect Youtube Comments and Perform Sentiment Analysis on it

Code:

./main.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from nltk.sentiment import SentimentIntensityAnalyzer
import json
import time
import nltk

nltk.download("vader_lexicon")

YOUTUBE_URL = "https://youtu.be/SM66GDRyIVY?si=W_LhZtTTDISMmmAF"

options = webdriver.ChromeOptions()
options.add_argument("--start-maximized")

driver = webdriver.Chrome(options=options)
wait = WebDriverWait(driver, 25)

driver.get(YOUTUBE_URL)

# Wait until comments section appears
wait.until(EC.presence_of_element_located((By.TAG_NAME, "ytd-comments"))))

def slow_scroll():
    """Scrolls very slowly to trigger YouTube lazy loading."""
    driver.execute_script("window.scrollTo(0, 400);")
    time.sleep(0.2)

def load_comments(target=200):
    current_count = 0
    last_count = -1
    stable_rounds = 0

    while current_count < target:
        for _ in range(10):
            slow_scroll()

            try:
                wait.until(
                    lambda d: len(d.find_elements(By.CSS_SELECTOR, "#content-text")) >
current_count
                )
            except:
                pass

            comments = driver.find_elements(By.CSS_SELECTOR, "#content-text")
            current_count = len(comments)
            print("Loaded comments:", current_count)
```

```

        if current_count == last_count:
            stable_rounds += 1
            if stable_rounds >= 5:
                print("\n△ No more comments loading. Stopping...")
                break
        else:
            stable_rounds = 0

        last_count = current_count

# Load comments
load_comments(200)

# Extract comment blocks
comment_elements = driver.find_elements(By.CSS_SELECTOR, "ytd-comment-thread-renderer")

comments_data = []
sia = SentimentIntensityAnalyzer()

for block in comment_elements[:500]:
    try:
        author = block.find_element(By.CSS_SELECTOR, "#author-text span").text.strip()
    except:
        author = ""

    try:
        text = block.find_element(By.CSS_SELECTOR, "#content-text").text.strip()
    except:
        text = ""

    # Sentiment calculation
    sentiment_score = sia.polarity_scores(text)

    if sentiment_score["compound"] > 0.2:
        sentiment = "Positive"
    elif sentiment_score["compound"] < -0.2:
        sentiment = "Negative"
    else:
        sentiment = "Neutral"

    comments_data.append({
        "author": author,
        "comment": text,
        "sentiment": sentiment,
        "scores": sentiment_score
    })

# Save JSON
with open("comments_with_sentiment.json", "w", encoding="utf-8") as f:
    json.dump(comments_data, f, indent=4, ensure_ascii=False)

print("\nFinal saved:", len(comments_data))
print("Saved to comments_with_sentiment.json")

driver.quit()

```

Output:

```
Welcome X scraping.py 6, U {} comments.json U X
{} comments.json > ...
1  [
2  {
3      "author": "",
4      "comment": "I appreciate all of you so much, thank you for helping do good in the world"
5  },
6  {
7      "author": "",
8      "comment": "This is beautiful and inspiring. Thank you."
9  },
10 {
11     "author": "",
12     "comment": "you are the best helper ever. helping 2,000 people is a world record."
13 },
14 {
15     "author": "@김지은-e3y4s",
16     "comment": "역시 믿고보는 미스터비스트\n2000명의 환자를 위해 의족을 선물하는 모습을 보고 눈물이 날뻔했습니다\n가족들까지 기뻐하고 눈물 흘리는 장면이 너무 마음아프네요ㅠ저도
17 },
18 {
19     "author": "@yunna271",
20     "comment": "The definition of wealth is held by the right person"
21 },
22 {
23     "author": "@pukarra1961",
24     "comment": "A single person is doing what the government could not do in years."
25 },
26 {
27     "author": "@CURSEDCHILD11.11",
28     "comment": "Helping and changing these many lives is inspiring !!"
```

```
Welcome X scraping.py 6, U {} comments_with_sentiment.json U X
{} comments_with_sentiment.json > {} 1 > {} scores > # compound
1  [
2  {
3      "author": "",
4      "comment": "I appreciate all of you so much, thank you for helping do good in the world",
5      "sentiment": "Positive",
6      "scores": {
7          "neg": 0.0,
8          "neu": 0.51,
9          "pos": 0.49,
10         "compound": 0.8617
11     }
12 },
13 {
14     "author": "",
15     "comment": "This is beautiful and inspiring. Thank you.",
16     "sentiment": "Positive",
17     "scores": {
18         "neg": 0.0,
19         "neu": 0.303,
20         "pos": 0.697,
21         "compound": 0.8481
22     }
23 },
24 {
25     "author": "",
26     "comment": "you are the best helper ever. helping 2,000 people is a world record.",
27     "sentiment": "Positive",
28     "scores": {
29         "neg": 0.0,
30         "neu": 0.506,
31         "pos": 0.494,
32         "compound": 0.8316
33     }
34 },
35 {
36     "author": "@김지은-e3y4s",
37     "comment": "역시 믿고보는 미스터비스트\n2000명의 환자를 위해 의족을 선물하는 모습을 보고 눈물이 날뻔했습니다\n가족들까지 기뻐하고 눈
38     "sentiment": "Neutral",
39     "scores": {
40         "neg": 0.0,
```

Answer the following questions based on above article.

1. Explain the Sentiment Analysis (SA) process pipeline.

The Sentiment Analysis process typically follows these steps:

1. **Data Collection**

Gathering data from social networks, APIs, web scraping, or existing datasets.

2. **Pre-processing**

Cleaning and preparing the text by removing:

- Stopwords
- Punctuation
- URLs
- Emojis (if required)
- Converting to lowercase, tokenization, stemming/lemmatization

3. **Feature Extraction**

Representing text using:

- Bag-of-Words (BoW)
- TF-IDF
- Sentiment lexicons
- Word embeddings (Word2Vec, GloVe, BERT vectors)

4. **Sentiment Classification**

Applying:

- Machine Learning models (SVM, Naive Bayes)
- Lexicon-based methods (VADER, SentiStrength)
- Deep learning models (LSTM, GRU, Transformers)

5. **Evaluation**

Measuring performance using:

- Accuracy
- Precision
- Recall
- F1-score

6. Visualization & Reporting

Presenting results using dashboards, charts, and insights for business decision-making.

2. List the generic and Social Network Sentiment Analysis Tools which are free to use.

Generic Free Sentiment Analysis Tools

- TextBlob
- VADER
- SentiStrength
- Stanford CoreNLP
- Pattern
- NLTK

3. List the tools that perform ER (Emotion Recognition) along with Sentiment Analysis in English.

Tools that support **Emotion Recognition (ER) + Sentiment Analysis**:

- **SentiStrength**
- **MeaningCloud**
- **Semantria**
- **TwinWord**
- **IBM Watson Tone Analyzer**
- **Text2Emotion (Python library)**

4. List the tools that allows URLs/ keywords/text/ as option for data entering for analysis and allows Jason/ csv as data export.

These tools allow multiple input types (URL, keywords, text) and support JSON/CSV export:

- MeaningCloud
- Social Mention
- Brand24 (limited free features)
- Semantria
- RapidMiner
- Google Cloud Natural Language API
- IBM Watson NLU

5. List the tools that support more than 5 text languages for the analysis.

Tools supporting multilingual (>5 languages) sentiment analysis:

- MeaningCloud
- IBM Watson NLU
- Google Cloud Natural Language
- TwinWord
- Semantria
- Rosette Text Analytics
- TextBlob (via Pattern multilingual support)

6. List the tools that support both APIs and client for the most used programming languages.

Tools offering APIs + SDKs for languages like Python, Java, JavaScript, PHP, etc.:

- Google Cloud Natural Language
- Microsoft Azure Text Analytics
- IBM Watson NLU
- MeaningCloud
- TwinWord
- Semantria

7. Write the concluding remark on the tool (among 24) that is best recommended for business analysis. Justify the same.

Recommended Tool: Google Cloud Natural Language API

Justification:

Google Cloud Natural Language API is the most suitable for business analysis because:

- It provides **highly accurate sentiment scoring** and **entity-level sentiment**, which is crucial for understanding customer opinions.
- Excellent **multi-language support**, scalable for global businesses.
- Offers **JSON export**, cloud integration, and enterprise-grade performance.
- Provides well-developed **APIs and client libraries** for major programming languages, making integration seamless.
- Ideal for handling **large volumes of data**, making it scalable for real-world business applications.